Harshit Rawal

AI / ML Engineer

Research interests: brain–computer interfaces · EEG signal processing · efficient AI on edge hardware

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Education

Sanjay Ghodawat University — Kolhapur, Maharashtra

B.Tech in Artificial Intelligence · CGPA: 7.36 / 10.0 — *Aug. 2022 – May 2026*

- **Relevant Coursework:** Machine Learning, Deep Learning, Neural Networks, Computer Vision, Natural Language Processing, Data Structures & Algorithms, Linear Algebra, Probability & Statistics, Basic Electrical & Electronics Engineering, Microprocessors and Microcontrollers.

Research Experience

Brain-Computer Interfaces & EEG Signal Decoding — Sanjay Ghodawat University

Undergraduate Research Project · Team of 4 — *2025 – 2026*

- Co-authored *"Review on Brain-Computer Interface and EEG Technologies"* (manuscript in preparation), a survey of EEG signal acquisition, preprocessing, and mental-state / motor-imagery decoding methods that formed the methodological basis for MindNav.
- Co-developed **MindNav**, a brain–computer interface that reads EEG signals in real time to drive a hands-free typing keyboard; worked on signal capture (Arduino + BioAmp), feature extraction (42 features), and the classical and deep-learning classifiers in PyTorch. github.com/Vasco1290/Mind-Nav

Experience

Autonex AI — Mumbai, India

Robotics Data Annotator / Reviewer — *Jul. 2026 – Oct. 2026*

- Annotated and quality-reviewed multi-camera surround-view autonomous-vehicle perception data across UK and US driving scenes, labeling road markings, vehicle classes, and driving maneuvers to produce high-quality ground-truth datasets for perception model training.
- Flagged ambiguous and edge-case labels and gaps in the labeling guidelines, improving dataset consistency.

VPN Digital Services Pvt. Ltd. — Mumbai, India

AI/ML Developer Intern — *Jan. 2026 – Jun. 2026*

- Built and deployed a production Android malware detection backend (PyTorch, LightGBM, ChromaDB) using a 7-stage pipeline that combines signature lookups, similarity search, and an ML ensemble; achieved **97.6% F1** on the TUANDROMD benchmark (7,130 apps).
- Built a phishing URL detection API (LightGBM, character n-grams) trained on 208K URLs; achieved **97.4% ROC-AUC** and cut false positives from 47 to 9 by rebalancing the data by domain.

Projects

Brokkr | *INT8 model compression & robustness benchmarking · In progress* — vasco1290.github.io/Brokkr

- Building an open benchmarking platform that compresses image classifiers to INT8 for edge hardware, stress-tests them under 12 real-world corruptions (darkness, fog, contrast and others from ImageNet-C), and publishes a per-model "nutrition label" covering size, accuracy, robustness, and whether the model's uncertainty signal still holds.
- Found that INT8 compression of MobileNetV3-Large costs only **1.98** top-1 points on clean images (95% interval −2.47 to −1.47) but **12–37** points under corruption (fog s3: −12.19; darkness s5: −13.58; contrast s5: −37.20), measured on the same 10,000 test images.

SecRAG | *Python, FastAPI, LangChain, ChromaDB, Next.js, Docker* — github.com/Vasco1290/SecRAG

- Retrieval-augmented question-answering system with cited, streaming answers; combines keyword and semantic search with a reranking model, and includes an automated evaluation suite that runs in CI.
- Dockerized full stack (FastAPI backend + Next.js frontend) with swappable LLM providers (Groq, OpenAI, Ollama) and background document ingestion.

Technical Skills

Languages: Python, Rust, C++, SQL
ML / DL: PyTorch, scikit-learn, LightGBM, MNE-Python, NumPy, Pandas, SciPy
LLM & RAG: LangChain, ChromaDB, HuggingFace Transformers
Backend & APIs: FastAPI, Flask, Firebase
Infrastructure: Docker, GitHub Actions (CI/CD), Linux, AWS, Microsoft Azure, Git
Signals & Embedded: EEG signal processing, Arduino, Raspberry Pi

Certifications & Awards

Microsoft Certified: Azure AI Fundamentals (AI-900)